# Technical Document #581: Blockchain - Comprehensive Overview

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Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

### Key Takeaways:

- Proof of Stake (PoS) selects validators based on their stake in the network.
- Smart contracts are self-executing contracts with terms directly written into code.
- Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself.